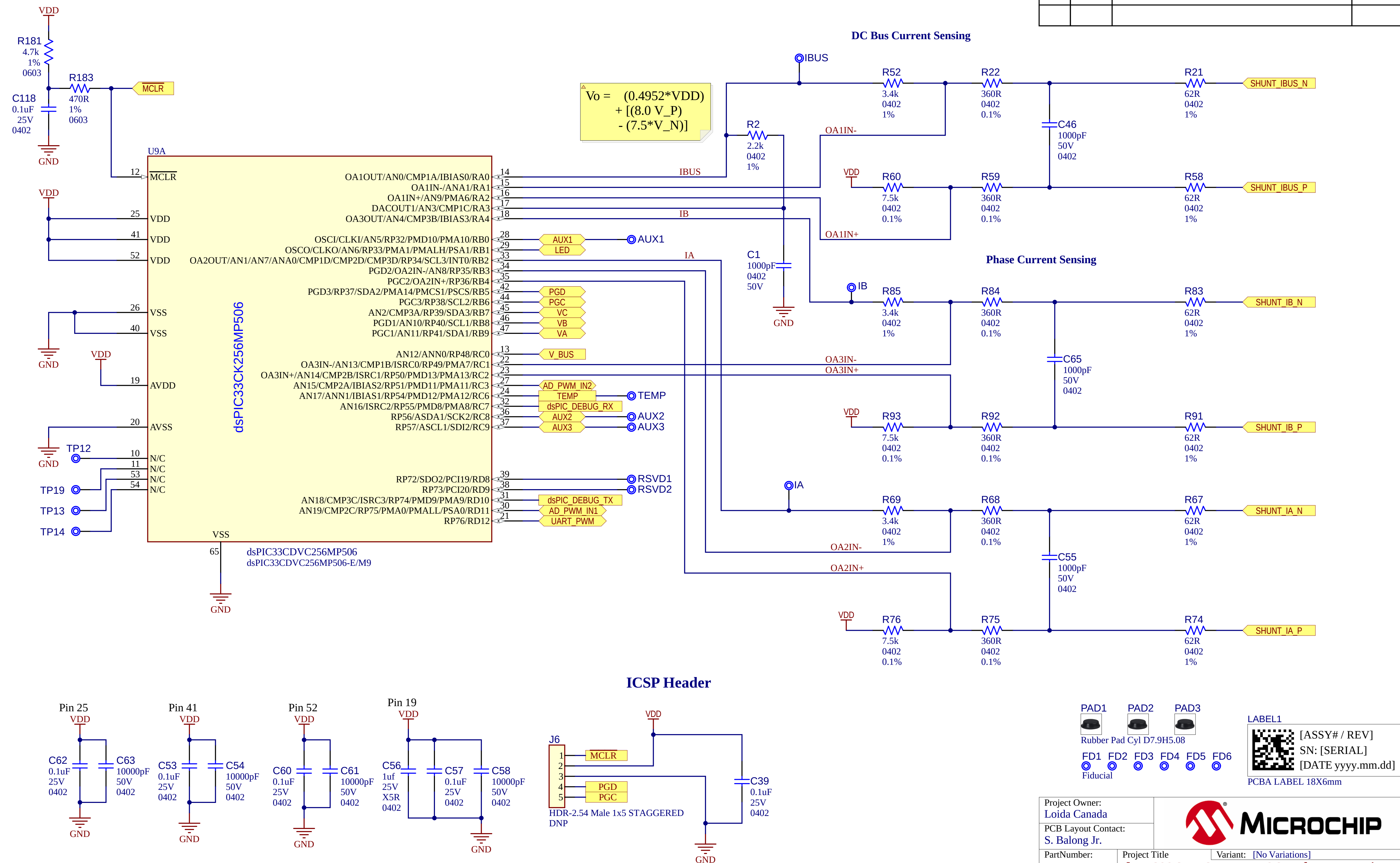


REV	ECO#	DESCRIPTION	DATE

Project Owner: Loida Canada

PCB Layout Contact: S. Balong Jr.

PartNumber: 12307

Project Title: dsPIC33CDVC Drone ESC Reference Design

Variant: [No Variations]

Sheet Title: Microcontroller and Internal OpAmp

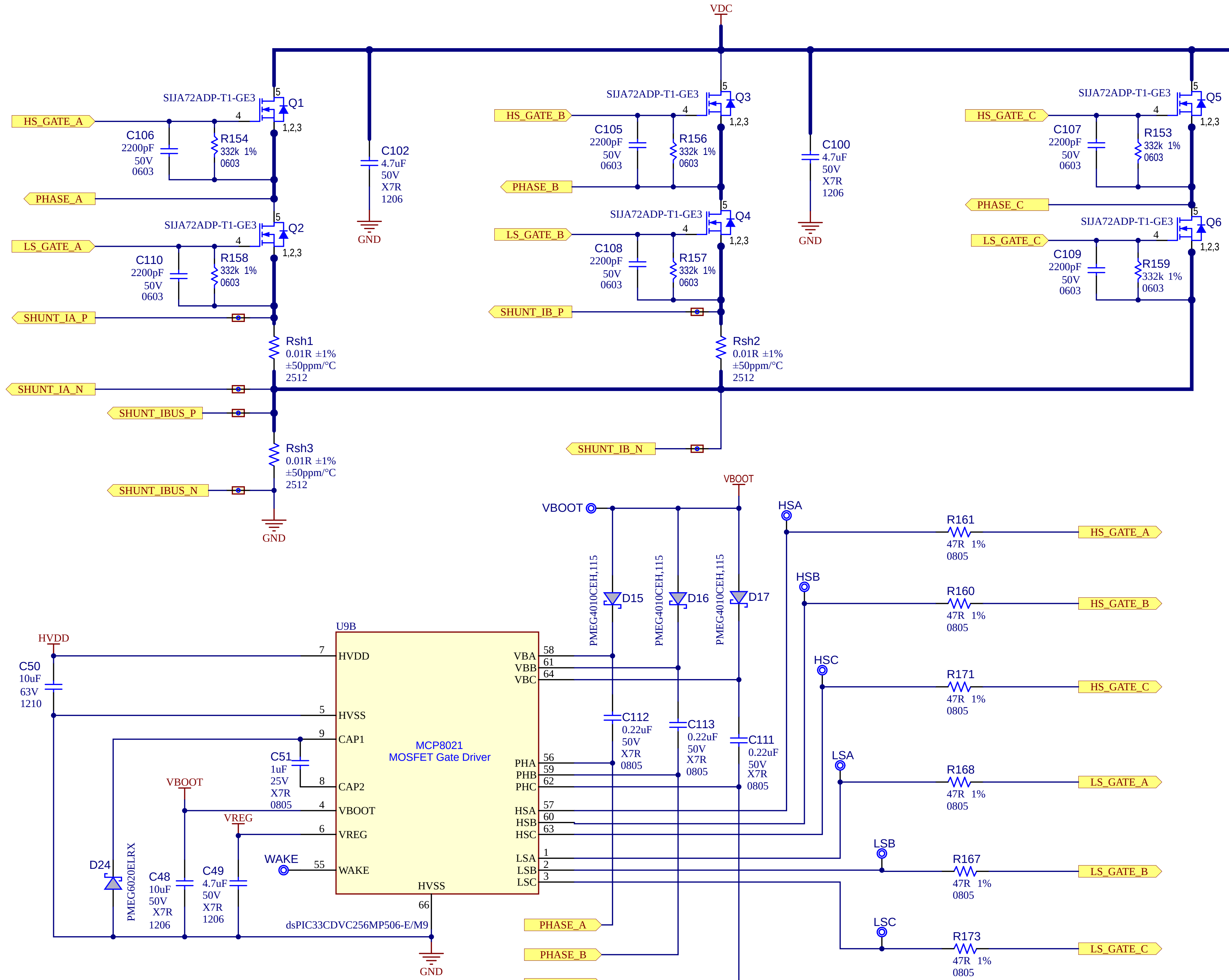
Size: B

SCH #: 03-01243 Rev: 2 Date: 2/13/2026

PCB #: 04-12307 Rev: 2 Sheet 1 of 5

File: Sheet1_dsPIC33CDV+OpAmp.SchDoc

Designed with Altium.com



PHASE CONNECTOR

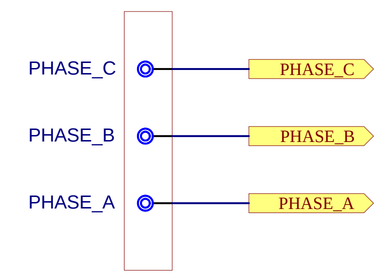


Table 1:

INTERNAL INTERCONNECTIONS dsPIC33CK256MP506 (U9A) to MOSFET Gate Driver (U9B)	
U9A	U9B
RP46/PWM1H/PMD5/RB14	PWMAH
RP47/PWM1L/PMD6/RB15	PWMAL
TMS/RP42/PWM3H/PMD1/RB10	PWMBH
TCK/RP43/PWM3L/PMD2/RB11	PWMBL
RP65/PWM4H/RD1	PWMCH
RP64/PWM4L/PMD0/RD0	PWMCL
TDI/RP44/PWM2H/PMD3/RB12	FAULT
RP60/PWM8H/PMD7/RC12	DE2
RP45/PWM2L/PMD4/RB13	OE

REV	ECO#	DESCRIPTION	DATE

Project Owner: Loida Canada
PCB Layout Contact: S. Balong Jr.
PartNumber: 12307
Project Title: dsPIC33CDVC Drone ESC Reference Design
Variant: [No Variations]
Sheet Title: Motor Control Inverter
Size: B
SCH #: 03-01243 Rev: 2
PCB #: 04-12307 Rev: 2
Date: 2/13/2026
Sheet 2 of 5
File: Sheet2_Inverter_GateDriver.SchDoc
Designed with Altium
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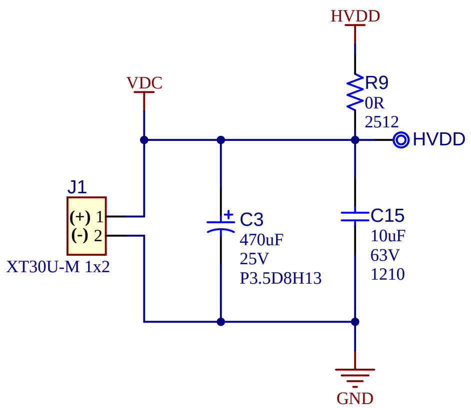
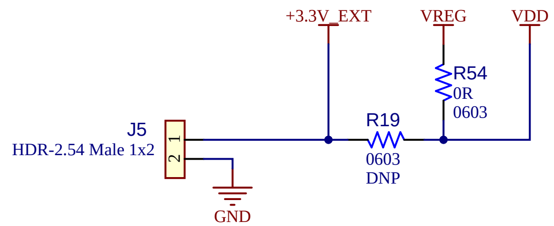
REV	ECO#	DESCRIPTION	DATE

Power Supply

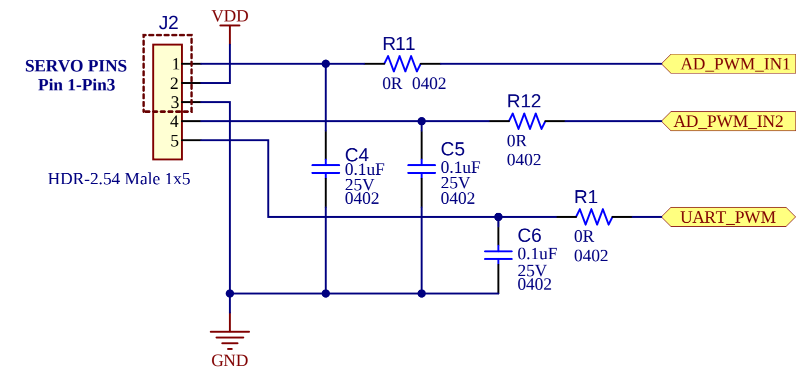




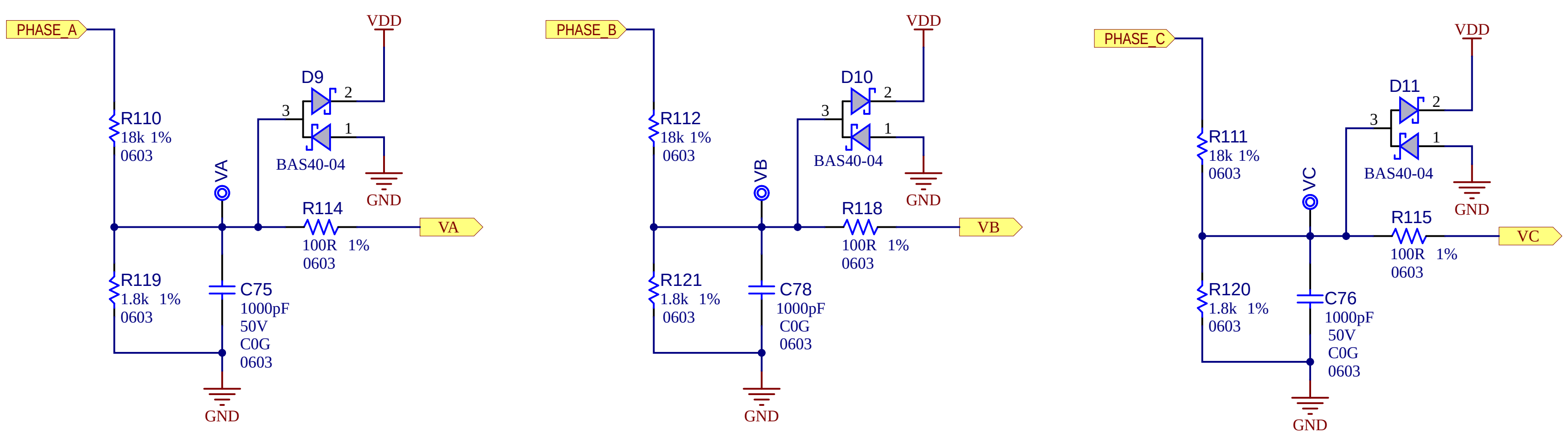
Ensure total power dissipation of the dsPIC33CDVC256MP506 (U9) does not exceed the specified value when internal 3.3V LDO (VREG- U9/U9B PIN#6) is used for powering dsPIC33CK256MP506 (U9A) by populating R54 instead of R19.

Resistor Configuration for the Supply Voltages:
VREG : R19=DNP and R54=0R, +3.3V is derived from Internal VREG (U9/U9B Pin #6)
: In this configuration, do not draw more than 20mA
+3.3V_EXT : When R19=0R, R54=DNP, 3.3V is derived from External Source

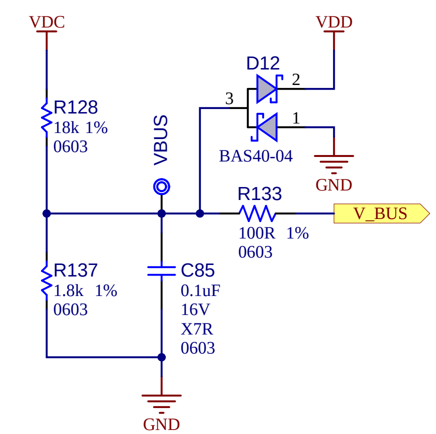
External PWM/ADC Control



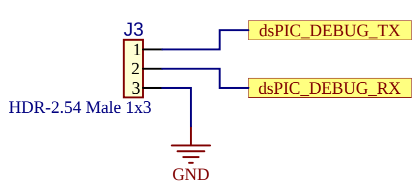
Phase Voltage Sensing



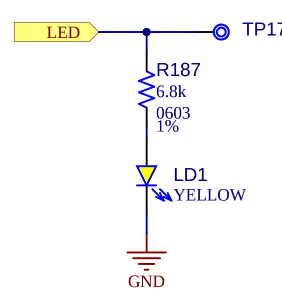
DC Bus Voltage Sensing



UART Communication Header



General Purpose LEDs



Project Owner: Loida Canada	MICROCHIP		
PCB Layout Contact: S. Balong Jr.			
PartNumber: 12307	Project Title: dsPIC33CDVC Drone ESC Reference Design	Variant: [No Variations]	
Sheet Title: Program/Debug Interface + Sensing/Feedback Ckts			Designed with Altium
Size: B	SCH #: 03-01243 / PCB #: 04-12307	Rev: 2 / Rev: 2	Date: 2/13/2026 / Sheet 3 of 5
File: Sheet3_SenseCkts_Programming_Interface.SchDoc			Altium.com

REV	ECO#	DESCRIPTION	DATE

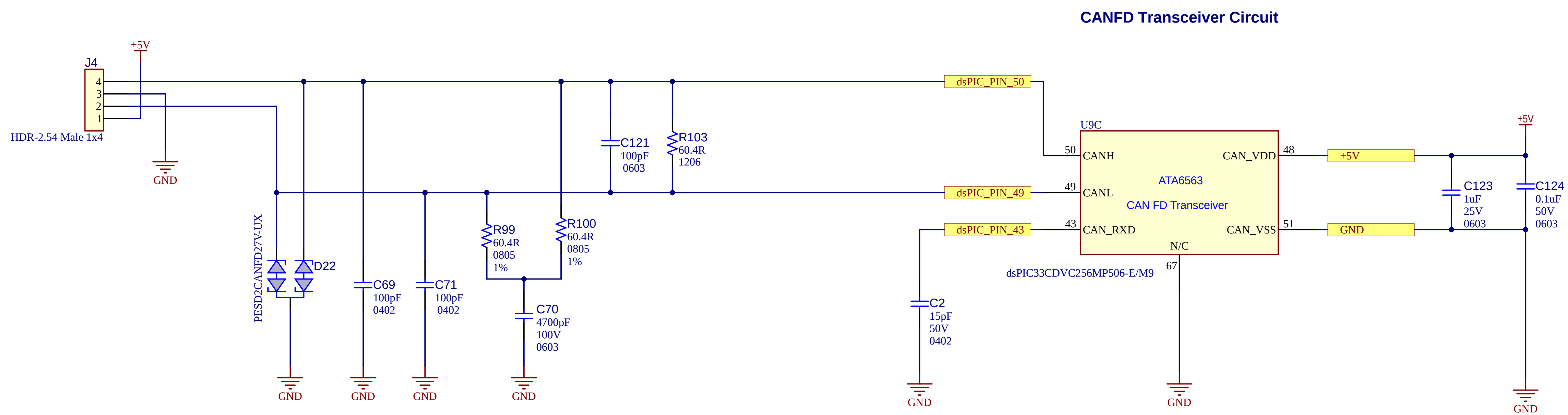


Table 2:

INTERNAL INTERCONNECTIONS dsPIC33CK256MP506(U9A) to CANFD Transciever(U9C)	
U9A	**U9C**
RP58/PWM7H/PMRD/PMWR/PSRD/RC10	**CAN_STBY**
RP59/PWM7L/RC11	**CAN_TXD**
RP52/PWM5H/ASDA2/RC4	**CAN_RXD**
VDD	**CAN_VIO**
Note : CAN_RXD is connected externally to the dsPIC33CDVC256MP506 PIN#43	

Project Owner:
Loida Canada
PCB Layout Contact:
S. Balong Jr.

MICROCHIP

PartNumber: 12307

Project Title: *dsPIC33CDVC Drone ESC Reference Design*

Variant: [No Variations]

Sheet Title: CAN Interface

Size: B | SCH #: 03-01243 | Rev: 2 | Date: 2/13/2026

PCB #: 04-12307 | Rev: 2 | Sheet 4 of 5

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File: Sheet4_CAN_Interface.SchDoc

REVISION HISTORY

REV 1

Initial Design

REV 2

> PCB Re-spin

> Changed PCB Material
from: 1oz All Layers,
to: 2oz External Layers / 1oz Internal Layers

> Changed Header Connectors from: 1.27mm pitch to: 2.54mm pitch

Project Owner: Loida Canada	MICROCHIP			
PCB Layout Contact: S. Balong Jr.				
PartNumber: 12307	Project Title: *dsPIC33CDVC Drone ESC Reference Design*	Variant: [No Variations]		
Sheet Title: Revision History				Designed with Altium
Size: B	SCH #: 02-01243	Rev: 2	Date: 2/13/2026	
	PCB #: 04-12307	Rev: 2	Sheet 5 of 5	Altium.com
File: Sheet5_RevHistory.SchDoc				